

Data Flow Diagram & User Stories

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Data Flow Diagram

The Level-1 DFD below represents the implemented data movement in PocketSmart AI.

External Entity / Store	Data Flow	Process
User	Registration/login data	1. Authentication & Request Handling
User	Budget, preferences, planner inputs	2. Planner Input Validation
2. Planner Input Validation	Validated structured input	3. Recommendation Orchestrator
3. Recommendation Orchestrator	Prompt + optional outfit image	Google Gemini API
Google Gemini API	Structured AI recommendation	3. Recommendation Orchestrator
3. Recommendation Orchestrator	Items, quantities, estimated unit prices	4. Python Budget Validation
4. Python Budget Validation	Validated plan + totals	5. Result Rendering
5. Result Rendering	Formatted recommendation	User
3/4. Services	Authenticated plan record	D1 SQLite Database
D1 SQLite Database	Saved history/dashboard data	User

User Stories

User Type	Functional Requirement	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Web User	Registration	USN-1	As a new user, I can register with account details.	Account is created with hashed password.	High	Sprint-1
Web User	Login	USN-2	As a user, I can log in and access my dashboard.	Valid credentials create an authenticated JWT cookie.	High	Sprint-1

Web User	Home Planner	USN-3	As a user, I can enter my home budget and requirements to receive an interior plan.	AI plan is generated and Python-validated within budget.	High	Sprint-2
Web User	Party Planner	USN-4	As a user, I can enter party details and selected services to receive a plan.	Recommendations reflect guest count, event and service choices.	High	Sprint-2
Web User	Jewelry Planner	USN-5	As a user, I can enter styling preferences and optionally upload an outfit image.	Gemini returns coordinated jewelry suggestions and outfit analysis when an image is supplied.	High	Sprint-3
Authenticated User	History	USN-6	As a user, I can view and delete my previous recommendations.	Saved plans are associated with my account and retrievable.	Medium	Sprint-3